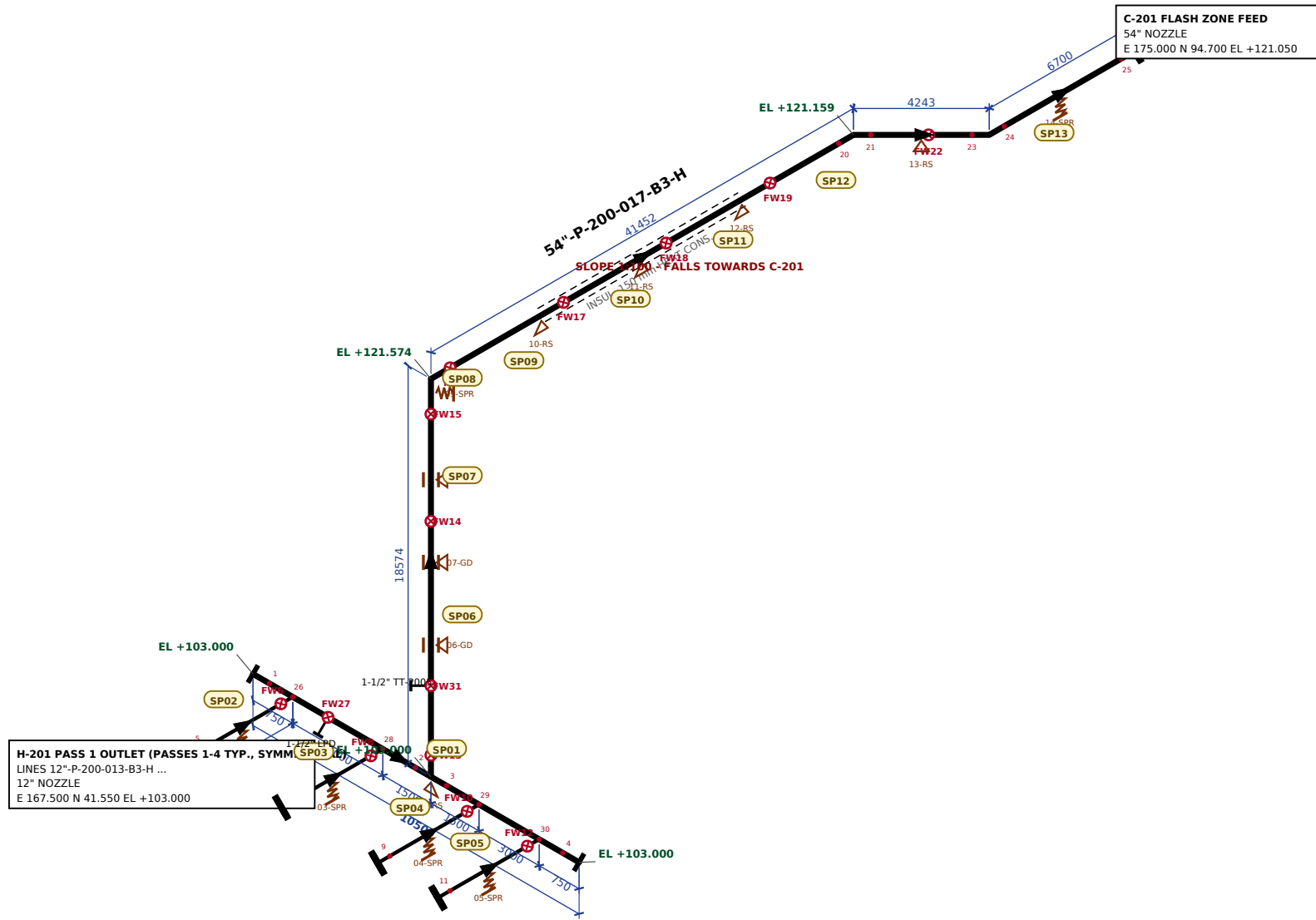
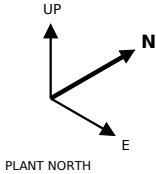


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



H-201 PASS 1 OUTLET (PASSES 1-4 TYP., SYMM
LINES 12"-P-200-013-B3-H ...
12" NOZZLE
E 167.500 N 41.550 EL +103.000

C-201 FLASH ZONE FEED
54" NOZZLE
E 175.000 N 94.700 EL +121.050

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	CAP, BW, WT 19.05, ASTM A234 WP9	54"	2
2	PIPE WT 19.05, BE, ASTM A691 9CR Cl.42 EFW	54"	69.4 m
3	TEE EQUAL, BW, WT 19.05, ASTM A234 WP9	54"	1
4	FLANGE WN RF CL300 B16.5, bore STD, ASTM A182 F9	12"	4
5	PIPE STD, BE, ASTM A335 P9 SMLS	12"	7.0 m
6	ELBOW 90 LR, BW, WT 19.05, ASTM A234 WP9	54"	1
7	ELBOW 45 LR, BW, WT 19.05, ASTM A234 WP9	54"	2
8	FLANGE WN RF CL300 B16.47A, bore WT 19.05, ASTM A182 F9	54"	1
9	WELDOLET (branch), ASTM A182 F9	12"	4
10	SOCKOLET 3000# (branch), ASTM A182 F9	1-1/2"	2
11	ERECTION: GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	12"	4
12	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	12"	4
13	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	54"	1
14	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	54"	1
15	PIPE SUPPORTS (std. dwgs): guide x3, rest x5, spring x6	-	14

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	54"	3636	SP01-2	54"	3636	SP02-1	12"	1751
SP03-1	12"	1751	SP04-1	12"	1751	SP05-1	12"	1751
SP06-1	54"	10600	SP07-1	54"	4917	SP09-1	54"	10600
SP10-1	54"	10600	SP11-1	54"	10600	SP12-1	54"	6743
SP12-2	54"	1450	SP13-1	54"	1089	SP13-2	54"	5568

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=10.5m SP02=1.9m SP03=1.9m SP04=1.9m SP05=1.9m SP06=10.6m SP07=4.9m SP08=4.1m SP09=10.6m SP10=10.6m
SP11=10.6m SP12=9.9m SP13=8.6m

NOTES

- Pipe/fittings/flanges per class B3 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 25.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Vacuum transfer line: riser at heater then continuous fall 1:100 to C-201 (no liquid hold-up).
- Large bore 54" for low velocity / flash-zone pressure; external-pressure (full vacuum) design.
- Crosses rack at high level on transfer-line support structure TLS-201 (independent of rack).
- Formal stress analysis mandatory; spring hangers on riser (heater + column growth).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 54"-P-200-017-B3-H

H-201 OUTLET VACUUM TRANSFER LINE TO C-201

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER	CFU-200-PI-ISO-001		
			REV
			A

LINE NUMBER	FROM	TO	PIPING CLASS
54"-P-200-017-B3-H	H-201 outlet manifold	C-201 flash zone	B3 - 300# 9Cr-1Mo ASTM A335 P9
DESIGN P / T	OPER. P / T	TEST P (HYDRO)	PIPE WALL
3.5 barg / 580.0 C	-0.7 barg / 398.4 C	25.2 barg (1.5xPxST/S=4.8)	WT 19.05 (19.05 mm) ext.P
INSULATION	PWHT	NDE	STRESS
H 150 mm	YES - all thk (P-No.5B, B31.3 Tbl	100 % RT of BW; 100 % MT/PT of fillets;	CAT 1 - T > 300 C & NPS >= 6; T > 200 C & NPS >= 1
P&ID	FLUID / PHASE	PAINT	SPOOLS / WELDS
CFU-200-PR-PID-012, CFU-200-PR-PID-013	P / M Vacuum transfer line	HT silicone/Al (insul.)	13 spools; 17 shop / 14 field